Defining Topological Quantum Field Theory

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1 Defining a Topological Quantum Field Theory

The goal of this chapter is to rigorously define the tools and objects one uses to capture the quantum system we are working with over a given manifold. To this end, the major inputs include the particle types and the manifold, and the output of our theory should define the Hilbert Space we work over along with the properties of understanding evolution under different operations.

Technically speaking, a formulation of a topological quantum field theory simply involves

$$\frac{\delta}{\delta g_{\mu\nu}}\langle \text{any correlator}\rangle = 0,$$

but this is more complex to work with and not as useful depending on what we are working with. Atiyah defined some axioms for any topological quantum field theory. I would like to first use the categorical language mentioned at the end of this chapter to give a very high-level abstract definition, then use the axioms as natural motivators as a result.

Definition 1 (Topological Quantum Field Theory (Abstract Version))**.** A **topological quantum field theory** is a functor $F : \mathbf{Cob}(D+1) \to \mathbf{Vect}$ where $\mathbf{Cob}(D+1)$ is the category of D-dimensional manifolds with morphisms as $D+1$-dimensional cobordisms and **Vect** is the category of vector spaces with morphisms as linear maps.

Atiyah formulated 4 axioms that follow the above definition. Let us define them here

Definition 2 (Topological Quantum Field Theory)**.** A **topological quantum field theory** satisfies the following axioms:

1. Given a D-dimensional manifold Σ, there is an associated Hilbert space $V(\Sigma)$.
2. Given two D-dimensional manifolds Σ_1 and Σ_2, then we have that

$$V(\Sigma_1 \sqcup \Sigma_2) = V(\Sigma_1) \otimes V(\Sigma_2)$$

3. If we are given that $\Sigma = \partial\mathcal{M}$ for Σ and $\mathcal{M}$ manifolds, then $Z(\mathcal{M}) \in V(\partial\mathcal{M})$
4. Let Σ^* denote the manifold Σ oriented in the reverse direction, then

$$V(\Sigma^*) = V^*(\Sigma)$$

where V^* is the dual space to V. In particular, this gives us a natural way to define bras and kets now. Suppose $\Sigma = \partial\mathcal{M} = (\partial\mathcal{M}')^*$, then we have that

$$Z(\mathcal{M} \cup_\Sigma \mathcal{M}') = \langle Z(\mathcal{M}')|Z(\mathcal{M})\rangle$$

where $A \cup_\Sigma B$ means that you glue ∂A to ∂B along Σ.

Remark 3. In particular, axiom 2 implies that $V(\emptyset) = \mathbb{C}$ since $V(\emptyset) = V(\emptyset) \otimes V(\emptyset)$. Thus, axiom 3 implies that for $\mathcal{M}$ a closed manifold, then $\partial\mathcal{M} = \emptyset$ so $Z(\mathcal{M}) \in \mathbb{C}$. Axiom 4 also gives us that

$$Z(\mathcal{M}^*) = Z(\mathcal{M})^\dagger$$

Definition 4 (Corbordism)**.** We have that Σ_1 and Σ_2 are **cobordant** if $\partial\mathcal{M} = \Sigma_1 \sqcup \Sigma_2$. Then, $\mathcal{M}$ is called a **cobordism** between Σ_1 and Σ_2.

2 Results of the Axioms

We now wish to describe some major properties of TQFT using this formalism that have been somewhat hinted to for the past couple of chapters.

First, we have that for $\mathcal{M}$ a cobordism between Σ_1^* and Σ_2, we have by the axioms that

$$Z(\mathcal{M}) \in V(\Sigma_1^*) \otimes V(\Sigma_2)$$

so that we can write it as

$$Z(\mathcal{M}) = \sum_{\alpha,\beta} U^{\alpha\beta} \left|\psi_{\Sigma_2,\alpha}\right\rangle \left\langle\psi_{\Sigma_1,\beta}\right|$$

with $\left|\psi_{\Sigma_2,\alpha}\right\rangle$ a basis for $V(\Sigma_2)$ and $\left\langle\psi_{\Sigma_1,\beta}\right|$ a basis for $V(\Sigma_1^*)$.

Example 5 (Identity Cobordism)**.** Consider the manifold $\mathcal{M} = \Sigma \times I$ for I a closed interval. We have that trivially $\partial\mathcal{M} = \Sigma \sqcup \Sigma^*$. Note that $\Sigma \times I$ is contractible to just Σ and $Z(\mathcal{M}) \in V(\Sigma^*) \otimes V(\Sigma)$ which are just the set of maps from V to V, we have that

$$Z(\Sigma \times I) = \sum_{\alpha} \left|\psi_{\Sigma,\alpha}\right\rangle \otimes \left\langle\psi_{\Sigma,\alpha}\right|$$

which is precisely the identity operator too, indeed satifying the above axioms.

Example 6 (Degeneracy of Ground States on a Manifold)**.** Quotienting out the endpoints of I, we get $\mathcal{M} = \Sigma \times S^1$. This gives us

$$Z(\Sigma \times S^1) = \mathrm{Tr}[Z(\Sigma \times I)] = \mathrm{Dim}[V(\Sigma)]$$

3 With Particles Now

The point of this section is to relate operations between manifolds and the world-lines to the Atiyah axioms to have a well-defined quantum mechanical theory. We now consider a joint systems with particles on a manifold Σ.

Suppose we have $\mathcal{M}$ such that $\partial\mathcal{M}$ is connected, with $\Sigma = \partial\mathcal{M}$. Axiom 3 gives the wavefunction at Σ as

$$\left|\psi\right\rangle = Z(\mathcal{M}) \in V(\partial\mathcal{M}).$$

Given a basis of $V(\partial\mathcal{M})$, we rewrite this as

$$\left|\psi\right\rangle = \sum_{\alpha} c_\alpha \left|\psi\right\rangle$$

Similar to before, we can consider $\mathcal{M}' \cup_\Sigma \mathcal{M}$ so that $\partial(\mathcal{M}' \cup_\Sigma \mathcal{M}) = \Sigma'$ with $\mathcal{M}'$ a cobordism between Σ^* and Σ', giving a time-evolved wavefunction

$$\left|\psi'\right\rangle = Z(\mathcal{M} \cup_\Sigma \mathcal{M}') \in V.$$

To understand the relation between these two wavefunctions, take a homeomorphism of Σ' so that the particles are aligned in the same relative positions as in Σ so that the Hilbert spaces are the same and thus share the same basis, so we have that

$$\left|\psi'\right\rangle = \sum_{\alpha} c'_\alpha \left|\psi_\alpha\right\rangle.$$

Additionally, since $\mathcal{M}'$ is a cobordism between Σ^* and Σ', we get that

$$Z(\mathcal{M}') = \sum_{\alpha,\beta} U^{\alpha\beta} \left|\psi_{\Sigma',\alpha}\right\rangle \left\langle\psi_{\Sigma,\beta}\right|.$$

This allows us to write a unitary transformation of the coefficients

$$c'_\alpha = \sum_\beta U^{\alpha\beta} c_\beta.$$

Ground-state degeneracy is the same process as before too, where

$$Z(\Sigma \times S^1) = \mathrm{Dim}[V(\Sigma)].$$

Remark 7 (Removing Particles). Start with a manifold $\mathcal{M}$ and world-lines L_i on the manifold. Cut out a tubular neighborhood around these world-lines (and assume these lines are closed loops), thus making a new manifold $\mathcal{M}'$ such that

$$\partial\mathcal{M}' = \partial\mathcal{M} \sqcup \left(\bigsqcup_i T_i^2 \right).$$

Suppose there are only two world-lines L_1 and L_2. We therefore can write by the axioms:

$$Z(\mathcal{M}) = \sum_{i,j} Z(\mathcal{M}; i, j) \langle \psi_{L_1,i}| \otimes \langle \psi_{L_2,j}|$$

where $Z(\mathcal{M}; i, j)$ is the partition function with particles i and j along L_1 and L_2 respectively. This makes sense since $Z(T^2)$ encodes the particle types in the Hilbert space. Thus, we have that $Z(\mathcal{M}; a, b)$ is simply $Z(\mathcal{M})$ glued with the wavefunction that satisfies the boundary condition giving $|\psi_{L_1,a}\rangle \otimes |\psi_{L_2,b}\rangle$.

4 On the S-matrix

Note that $S^3 = (S^1 \times D^2) \cup_{T^2} (D^2 \times S^1)$ so we have that

$$Z(S^3) = \big\langle Z(S^1 \times D^2) \big| Z(D^2 \times S^1) \big\rangle$$

immediately from the axioms.

Definition 8 (Modular S-matrix). Since $D^2 \times S^1$ is a solid torus, consider world-lines of two different particles on each of the solid torii on this S^3. This gives the following:

$$Z(S^3; a \text{ loop linking } b \text{ loop}) = \big\langle Z(S^1 \times D^2; b) \big| Z(D^2 \times S^1; a) \big\rangle \equiv S_{ab}$$

where S_{ab} is the **modular S-matrix**.

We can do a similar construction with $S^2 \times S^1 = (D^2 \times S^1) \cup_{T^2} (D^2 \times S^1)$ to get that

$$\big\langle Z(D^2 \times S^1; b) \big| Z(D^2 \times S^1; a) \big\rangle = \delta_{ab}$$

since the basis is the same for both cases now. Thus,

$$\big\langle Z(D^2 \times S^1) \big| Z(D^2 \times S^1) \big\rangle = \delta_{II} = 1.$$

Thus, by the axioms, gives us that

$$Z(S^2 \times S^1) = 1$$

which we knew was true, since

$$Z(S^2 \times S^1) = \mathrm{Dim}(V(S^2)) = 1.$$